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Asphaltene Deposition in Porous Media Micromodels: Experimental Studies and Comprehensive Permeability-Reducing Mechanisms

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Highlights

- The mechanism of asphaltene deposition and fouling in porous media are studied;
- Asphaltene deposits and permeability loss in porous media are investigated;
- The impact of the different parameters on asphaltene accumulation is discussed.

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Abstract

Asphaltene-induced formation damage is one of the complicated processes of permeability damage in porous media, particularly in the near-wellbore area. Asphaltene particles precipitate out of the bulk fluid phase during production due to pressure drop, which may reduce permeability owing to the deposition of asphaltene nanoparticles on porous media surfaces and the plugging of pore throats by larger asphaltene agglomerates. Asphaltene precipitation and deposition in production tubes and surface facilities are well-documented concerns, and many solutions for managing them are available. However, the effects of asphaltene in the reservoir, particularly in the near-wellbore zone, are little known. In this study, using an artificial porous medium, experimental data on pressure drop due to changes in parameters such as the flow rate, the type of precipitant n-alkane solvent (n-heptane alkane solvent and n-decane), and the percentage of precipitant are obtained. Next, the permeability reduction from asphaltene deposition in a porous medium is calculated. Experimental data are fitted with the proposed quasi-experimental models at different time intervals to identify the dominant mechanism in reducing clogging. One of the study's accomplishments is determining the principal mechanism of permeability reduction (in vitro) using a reasonably basic model with the least dependent parameters and a decent approximation. According to the findings, pore throat plugging becomes the dominant mechanism of permeability reduction although filtration cake formation and surface deposition may exist during the tests.

Keywords: Asphaltene deposition, Glass micromodel, Permeability damage, Porous media, Solvent injection.

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1. Introduction

Deposition of asphaltene can cause challenges in both upstream and downstream processes (Hamadou et al., 2008). Asphaltene precipitation may accumulate in pipelines, manufacturing facilities, and heat exchangers in downstream operations, which is common when the Reynolds number is high. A great deal of research is now being conducted to address this issue. Asphaltene precipitates may settle in the porous matrix in upstream operations, leading to decreased porosity and permeability and changes in wettability (Mohammed et al., 2021). Asphaltene adsorbs on the pore surface, covering it and changing its wettability owing to electrochemical interactions. Polar components of crude oil are considered the primary cause of adsorption.

Furthermore, the rate of adsorption by polar molecules is proportional to the wettability change. These phenomena often occur at low Reynolds numbers and are poorly understood. To accurately predict the rate of permeability damage in reservoirs caused by asphaltene deposition and devise an appropriate cleanup treatment, it is necessary to accurately forecast the rate of permeability damage in reservoirs caused by asphaltene deposition and apply the necessary remediation treatment. Two factors must be examined to estimate permeability damage. The first is the amount of asphaltene that deposits in porous media, and the second is the measure of permeability loss associated with a particular amount of asphaltene deposition. This work uses laboratory experiments to answer the latter question. The oil industry prioritizes understanding the asphaltene accumulation problem and the variables that influence it due to the apparent high cost involved with production loss because of asphaltene deposition and remediation techniques such as solvent wash and removal of deposited asphaltene (Soulgani et al., 2011). There is well-documented research available in the literature on the latter case. Several experimental approaches and modeling tools have been developed during the last decade to explore and monitor asphaltene deposition in flow lines (Hassanpouryouzband et al., 2017; Ghahfarokhi et al., 2017; Naseri et al., 2020; Guan et al., 2017). However, the difficulties associated with asphaltene fouling inside reservoirs are even less well-known owing to a lack of experimental capabilities.

Moreover, the present modeling techniques for evaluating asphaltene deposition in porous media have limited capabilities, either due to a lack of reliable lab data for validation or to the involvement of too many changeable factors that cannot be confirmed or quantified in the lab. Numerous modeling methodologies are utilized in the literature to model asphaltene deposition in porous media (Gruesbeck et al., 1982; Kord et al., 2014; Eskin et al., 2016). Several mechanisms, such as adsorption, surface deposition, pore throat plugging, deposit entrainment, and pore throat opening, have been examined in these techniques. However, it is still unclear which mechanisms are essential and which could be ignored in modeling asphaltene deposition inside porous reservoir rocks.

The pore-scale investigation of asphaltene accumulation in porous media is an experimental technique that aids in comprehending the mechanics of asphaltene deposition (Blunt et al., 2002). Researchers can explore the asphaltene accumulation profile in porous media using visual observation and image processing technologies (Fadili et al., 2009; Lin et al., 2016). Microfluidic devices can also analyze the flow pattern near asphaltene deposits. Several experiments have been performed to examine asphaltene deposition as follows:

Bagheri et al. investigated the influence of various production schemes, such as natural depletion and CO₂ injection, on the propensity of asphaltene accumulation in porous media (Pak et al., 2011). Jafari Behbahani et al. (2012) used a bottom-hole live oil sample to examine the permeability decline of a tight-sandstone core sample under miscible CO₂ flooding procedures induced by asphaltene accumulation. Leontaritis (1998) constructed an asphaltene near-wellbore formation damage method to evaluate the severity of formation damage caused by asphaltene accumulation while including time and

the impact of asphaltene deposition on the hydraulics of the near-wellbore region as well as the wellbore itself. Shirdel et al. (2012) investigated asphaltene deposition using the spherical particle model and varied deposition methods based on size. Their results revealed that the deposition rate was reduced by increasing flow velocity due to expansion and decreasing tubing surface temperature due to the thermal gradient from bottom-hole to surface. Mirzabozorg et al. (2009) investigated permeability variations caused by asphaltene deposition using a dual-porosity, dual-permeability model and found that asphaltene deposition had a more significant influence on fracture permeability than matrix deposition.

This study expands on the experience of the glass micromodel and extends the research to two precipitants such as n-heptane and n-decane, as well as the flow rate and the solvent-to-oil ratio. Asphaltene deposition is tracked by photographing the glass micromodel and monitoring the pressure decrease across it. This paper proposes a novel, simple model for a coupling pressure drop to decrease permeability. According to the model, conductivity changes are related to the square root of the pressure decrease. Furthermore, experimental data are adapted to the provided models for mechanisms such as surface deposition, pore throat plugging, and filtration cake formation, and the coefficients of the models are calculated to determine the dominating mechanism in reducing permeability. The physics of particle deposition in a porous media will be described in the subsequent sections. Improved models to identify deposition mechanisms and permeability impairment in porous media are still challenging due to numerous types of crude oil with different forms of asphaltene particles and diverse interactions of asphaltene with other particles in crude oil. Furthermore, permeability degradation caused by asphaltene precipitation may be influenced by medium porous properties and flow hydrodynamics.

2. Physics of permeability reduction

Roque et al. (1995) conducted extensive tests on porous quartzite stone and accurately described the sample before measurements. The influence of several factors on the permeability of the porous medium was explored, including the operational parameters, flow rate, concentration, and particle size. The obtained result showed the importance of such factors, particularly in light of the underlying condition that precipitation must be considered. Figure 1 shows the results in brief. Particle location and deposition kinetics are essential for considering permeability decline and deposition mechanisms. Further research has revealed that permeability decrease largely relies on the mechanism of particular particle deposition.

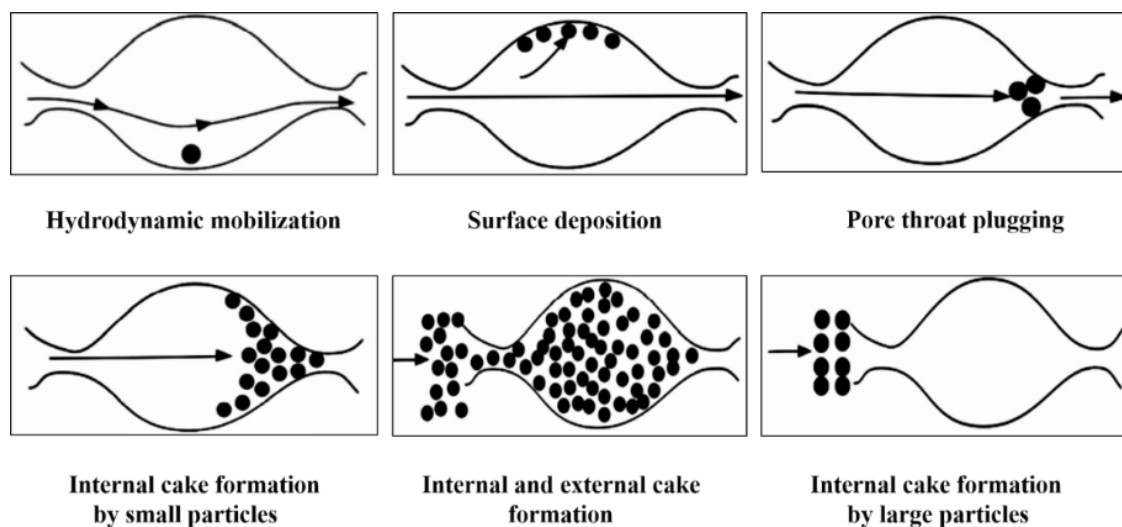


Figure 1

Various particulate processes (Roque et al., 1995).

2.1. Particle surface deposition

The deposition location in a surface deposition is the pore surface. The electrostatic charge differential between the surface of the pore and the particle, as well as the structure of the pore surface, chemical characteristics, and pore density, all impact the kinetics of the deposition process on the pore surface. However, if the deposition on the pore surface begins near the pore throat, permeability is considerably reduced. As a result, the impact of throat sediment fractions on permeability decrease is more important than the amount of sediment present. If the suspension flow velocity remains constant, the surface deposition becomes strictly monolayer due to tremendous repulsion between almost static particles. For this phase, the monolayer deposition occurs if the negative energy of aggregation is significant. From experimental and theoretical results, it can be concluded that the deposition on the surface has a minor effect on permeability reduction. Small-sized particles (e.g., clay-size and colloidal) are the favorite species to follow the surface deposition mechanism as the large particles are not available (Blauch et al., 1999).

2.2. Pore-throat plugging by bridging

Pore-throat bridging occurs when a particle goes through a pore throat and forms a bridge. The particle can link to two previously deposited particles on a pore-throat surface (three particles bridging) or to a previously deposited particle and the pore-throat area (two particles bridging). When particles are more significant than the pore-throat size, pore-throat bridging can happen (single-particle bridging). The pore bridge gives a means for upstream particle collection, significantly decreasing the fluid flow velocity via these pores. During this phase, the most rapid decrease in permeability is observed (Civan, 2016 and 2011).

2.3. Internal cake formation

When the proportion of bridged pore throats approaches a critical level, the pores are no longer connected across some critical damage layer. When the proportion of bridged pore throats hits a certain level, the pores are no longer connected across some critical damage depth. A fast decline in downstream particle concentration signifies the beginning of internal cake formation. The distribution and abundance of smaller particles, including colloidal and clay-sized, in the flow stream measure the extent of permeability damage (Manshad et al., 2018).

2.4. External cake formation

Particles accumulate upstream as internal cake formation is completed and may form filter cakes at the interface. This process is unimportant in a fracture/gravel-pack system because internal filter-cake formation dominates and continues over the length of the fracture (Blauch et al., 1999).

3. Model for permeability reduction

A simple model was created to describe the impact of different variables on asphaltene deposition and to determine a relationship between capillary tube deposition and formation damage (Bemani et al., 2017). The friction pressure drop (P) is determined using Poiseuille's laminar flow equation (Bemani et al., 2017).

$$V = \frac{\Delta P D^2}{32 \mu L} \quad (1)$$

where L indicates the length of the tube over which pressure loss is recorded (cm), D is the diameter of the capillary (cm), V denotes the fluid velocity (cm/s²), μ represents fluid viscosity (cP), and P is pressure loss over length L (dyne/cm).

The flow velocity is sub-stationed in the previous equation as Equation (2) (Bemani et al., 2017), where q is the volume rate of flow (cm³/s), giving Poiseuille's equation for laminar flow Equation (3) (Bemani et al., 2017):

$$V = \frac{4q}{\pi D^2} \quad (2)$$

$$\Delta P = \frac{128L\mu q}{\pi D^4} \quad (3)$$

The following equation calculates the initial pressure drop (Bemani et al., 2017).

$$\Delta P_0 = \frac{128L\mu q}{\pi D_0^4} \quad (4)$$

where ΔP_0 and D_0 are the initial pressure loss and capillary diameter, respectively. As a result, the equation is used to compute the ratio of pressure decrease at a certain period to the beginning pressure drop (Bemani et al., 2017):

$$\frac{\Delta P}{\Delta P_0} = \left(\frac{D_0}{D} \right)^4 = \left(\frac{r_0}{r} \right)^4 \quad (5)$$

Making a connection with Darcy's formula for linear flow in a porous medium (Equation (6)), permeability can be equated with the following (Bemani et al., 2013):

$$q = \frac{kA\Delta P}{\mu L} \quad (6)$$

$$k = \frac{r^2}{8} \quad (7)$$

The permeability ratio derived from Equations (6) and (7) is expressed by the following (Bemani, Ashoori, and Bahrami, 2017):

$$\frac{K}{K_i} = \sqrt{\frac{\Delta P_i}{\Delta P_t}} \quad (8)$$

where ΔP_t depicts the current pressure drop, and K/K_i represents the permeability ratio for a moment and indicates permeability reduction.

Wojtanowicz et al. (1987) proposed a theory of solid particle filtration that provides a tool for identifying permeability impairment processes such as surface deposition, pore throat blockage, and filter cake development. They suggested that Equations (9)–(11) describe permeability decrease via the method mentioned above:

$$\sqrt{\frac{K}{K_0}} = 1 - A \times t \quad (9)$$

$$\frac{K}{K_0} = 1 - B \quad (10)$$

$$\frac{K}{K_0} = \frac{1}{1 + C \times t} \quad (11)$$

where K indicates permeability at time t , K_0 represents reference permeability, t denotes time, and A , B , and C are constants in the dimension of the inverse of time.

It should be mentioned that these mechanisms are investigated under conditions in which only one permeability damage mechanism is prominent; nevertheless, in most situations, more mechanisms may coexist. Bagherzadeh et al. introduced a model combining surface deposition and throat-blocking models to evaluate the coexistence of two processes simultaneously (Qian et al., 2019). This research group experimented on formation damage due to asphaltene precipitation under CO_2 injection conditions in one of the Iranian carbonate oil reservoirs and developed an experimental model. They demonstrated that when surface deposition and pore-plugging processes coexisted, Equation (12) could accurately represent permeability behavior:

$$\frac{K}{K_0} = a \times t^2 + b \times t + c \quad (12)$$

Furthermore, excessive solid concentrations at the pore surface might form a cake layer, which limits the flow channel and reduces permeability. As a result, Equation (11) is changed to alter the moment the filter cake production process begins. This time is denoted by t_0 , and the solution to Equation (13) is as follows:

$$\frac{K}{K_0} = \frac{1}{1 + C \times (t - t_0)} \quad (13)$$

4. Experimental procedures

Experiments on asphaltene accumulation in porous media might be divided depending on how the oil and solvent are blended: outside (external mixing) or in the porous medium (internal mixing). Another method is to dissolve asphaltene particles in a solvent (such as hexane) and inject the mixture into porous media. The three methods are discussed below, assuming we eventually choose the internal mixing strategy.

4.1. Asphaltene particle suspension injection in porous media

Asphaltene particles are suspended in solvents, such as hexane, decane, or heptane, and injected at a constant flow rate into a new porous media, a standard methodology used to study formation damage caused by asphaltene deposition. Using this approach, keeping track of the amount of asphaltene that has entered the porous medium is also straightforward.

4.2. Oil/solvent mixture injection in porous media

In this procedure, one syringe pump injects oil while another injects solvent. The laboratory setup may inject oil and solvent into a mixer at widely varying but consistent flow rates linked to a glass micromodel. The solvent quantity in the injected mixture varies as the solvent-to-oil flow rate ratio changes. The pressure difference along the porous medium is measured, and while the flow rate and viscosity of the injected fluids remain constant, any variation in the pressure difference reveals changes in the permeability of the porous media. Figure 2 depicts a schematic of the micro model setup used in this study. Two constant-rate programmable syringe pumps (BS-8000/9000 from Braintree Scientific, the USA.), a glass micromodel, a camera, a pressure transducer well-calibrated in a range of 1.5×10^5 Pa, a data logger, and an accumulator are all components of this setup.

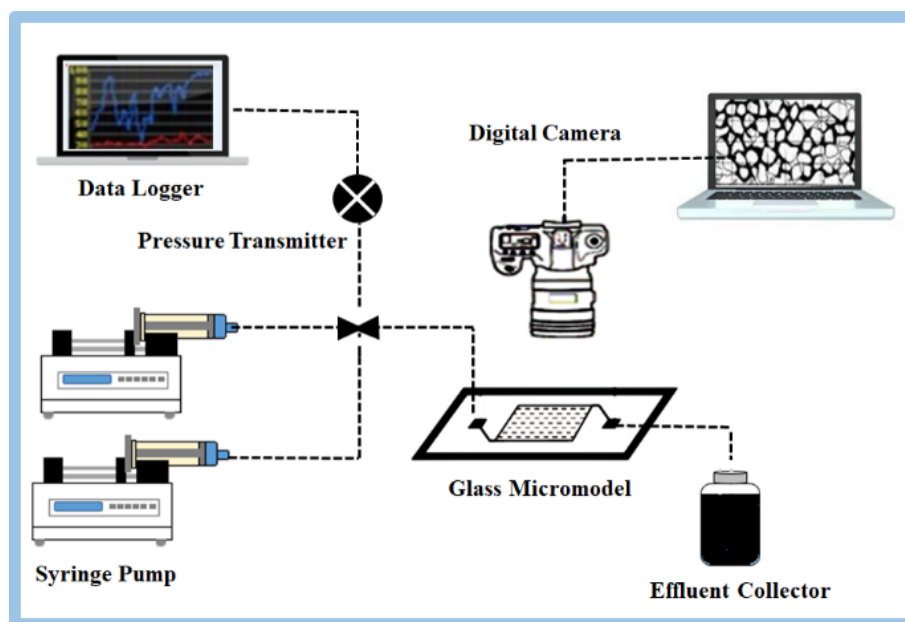


Figure 1

The setup of the micromodel experiment and visualization system.

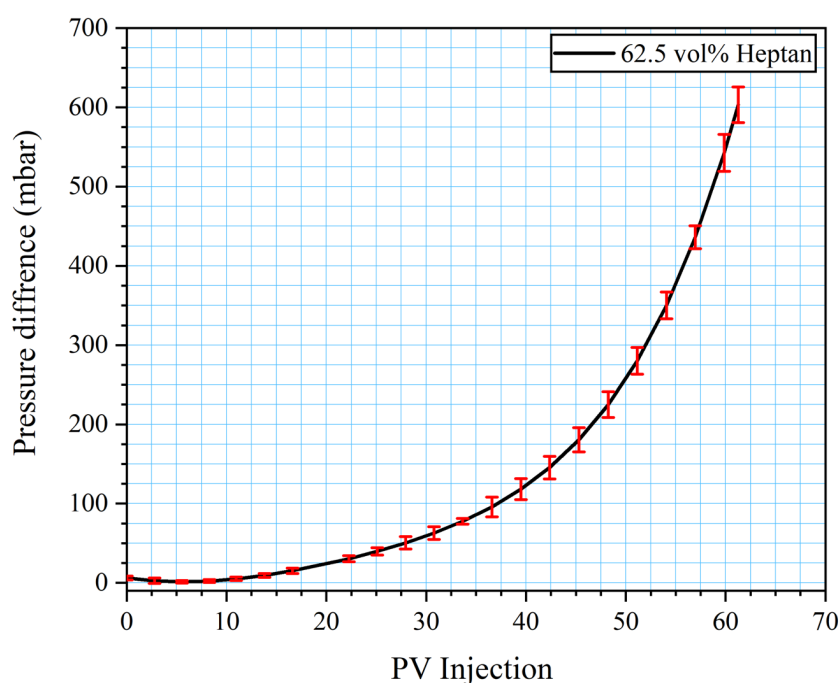
This study uses a sample of crude oil from one of the fields in southern Iran. n-decane and n-heptane are used as precipitants. The inlet fluid of crude oil and precipitant enter the micro-glass model after the combination, and the inlet and outlet pressure difference of the glass micromodel is recorded at this time. Over time, the pressure difference increases due to asphaltene deposition, and the experiment continues until the micro model is completely clogged. Tables 1 and 2 list the physical properties of the micromodels and the compositions of the crude oil samples. It should be emphasized that each experiment in this study is conducted twice, and the average results are reported. Figure 3 shows the error bar diagram (inlet fluid containing 62.5% by volume of n-heptane and a flow rate of 8 ml/h). The experimental data are also smoothed for easier comparison.

It should be noted that this set of tests is performed at laboratory temperature and pressure. On the other hand, a precipitate is used to destabilize the asphaltene. However, the temperature and pressure of oil tanks are higher than the laboratory temperature and pressure and are formed in oil fields due to changes in temperature and pressure of asphaltene deposition. This difference in conditions can significantly impact the results. The closer the test conditions to the oil reservoir conditions, the more reliable the results obtained. However, performing such tests in the laboratory can be less cost-effective in understanding the mechanisms of sediment formation and creating a scenario to prevent it.

Table 1

The physical properties of the micromodel

Size of micromodel	Throat length	Throat diameter	Number of pores in the width	Number of pores in the length	Coordination number
$10 \times 5 \text{ cm}^2$	0.5 mm	1.6 mm	32	108	6

**Figure 3**

The error bar diagram of pressure drop versus time, precipitant to oil ratio (62.5% v/v), and constant total flow rate ($Q_t = 8 \text{ mL/h}$).

Table 1

The composition and characteristics of the crude oil

Specification	Unit	Value
The density of dead oil	°API	24.9
Viscosity at ambient temp	cP	2.21
Asphaltenes	Mass fraction	0.041
N ₂	Mole fraction	0.000
CO ₂	Mole fraction	0.000

Specification	Unit	Value
H ₂ S	Mole fraction	0.000
C ₁	Mole fraction	0.000
C ₂	Mole fraction	0.319
C ₃	Mole fraction	0.938
i-C ₄	Mole fraction	0.431
n-C ₄	Mole fraction	1.896
i-C ₅	Mole fraction	1.630
n-C ₅	Mole fraction	2.775
C ₆	Mole fraction	6.363
C ₇	Mole fraction	8.300
C ₈	Mole fraction	9.541
C ₉	Mole fraction	8.282
C ₁₀	Mole fraction	7.422
C ₁₁	Mole fraction	5.961
C ₁₂	Mole fraction	4.828
C ₁₃	Mole fraction	4.473
C ₁₄₊	Mole fraction	37.734

5. Experimental results

5.1. Effect of precipitant type on asphaltene deposition

Asphaltene is a polydisperse combination of the most polarizable and heaviest components of the oil. Applying various n-alkane precipitants leads to precipitates with considerable changes in chemical and physical characteristics. Tavakkoli et al. (2014) used an indirect technique to demonstrate that lighter hydrocarbons were stronger asphaltene precipitants. According to research, the precipitated asphaltene increased as the carbon number of the precipitant dropped, implying that lighter hydrocarbons were stronger precipitants. Thus, the volume fraction of the precipitant required to cause asphaltene precipitation increased as the carbon number of the precipitant rose. As lighter hydrocarbon is used, the lighter components in crude oil increase and the polarity of crude oil decreases. Asphaltene, the heaviest fraction of the oil, is insoluble in light hydrocarbons; therefore, the oil becomes a poor solvent for the asphaltene, and more asphaltene is expected to precipitate when lighter hydrocarbons are used as precipitants. The onset of asphaltene precipitation increases as the carbon number of the precipitant increases. This change in onset volume fraction is related to the Gibbs free energy of mixing, which is determined by the difference in solubility parameters between n-paraffin and asphaltene, as well as the ratio of the asphaltene molar volume to n-paraffin molar volume. As the carbon number of n-paraffin increases, the solubility parameter difference decreases, and the molar volume ratio decreases in the entropy of mixing. Both changes reduce the Gibbs free energy of mixing and improve the compatibility of asphaltene and precipitant. It is also reported that asphaltene precipitated by stronger precipitants is heavier and more aromatic than asphaltene precipitated by lighter hydrocarbons (Tavakkoli et al., 2014). Figure 4 shows the micro-model pressure drop in the experiment with n-heptane and n-decane. According to the results of n-heptane injection, it has caused more asphaltene deposition and more pressure drop.

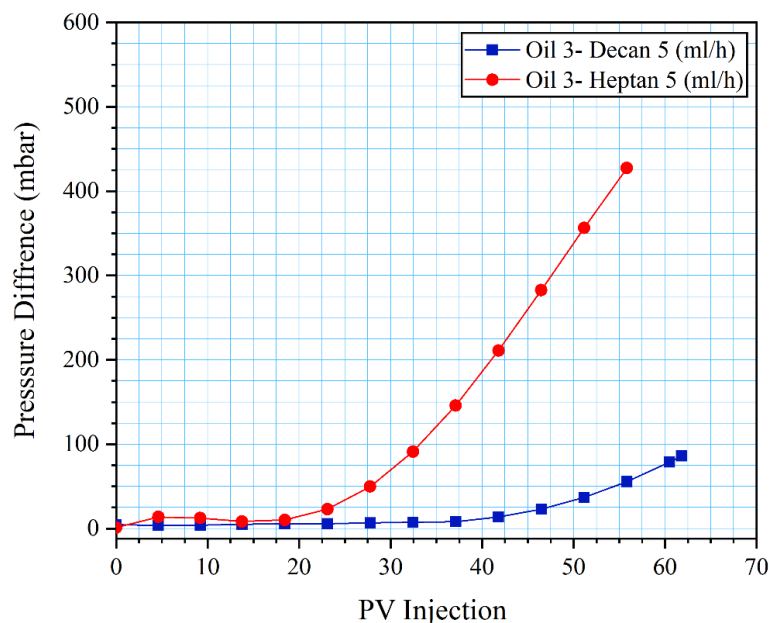


Figure 4

Pressure drops versus time for a constant precipitant-to-oil ratio (62.5% v/v) and a constant total flow rate ($Q_t = 8$ mL/h) for n-heptane and n-decane.

5.2. Effect of the flow velocity on asphaltene deposition

Figure 5 shows the effect of the flow velocity on the deposition behavior. The flow rate varies from 4 to 12 mL/h at a constant n-heptane-to-oil ratio. According to Figure 5, the deposition rate rises as the flow rate increases and shows a higher absolute pressure. This indicates that at higher flow rates, more asphaltenes deposit with significantly faster deposition kinetics. According to Nabzar and Aguilera (Nabzar et al., 2008), the deposition kinetics reduces as the flow rate increases due to entrainment or detachment. This is directly opposed to the findings of this study, and the difference in rates used can explain this contrast.

Furthermore, the observed contrast can be attributed to a balance between deposition kinetics and shear removal. Shear removal plays a vital role in deposition over a specific shear rate, potentially resulting in a reduced deposition rate with a rising flow rate. The result of the increasing mass of deposited asphaltenes with increasing flow rates suggests that the velocities utilized are in the domain where deposition kinetics overcomes shear removal. More experiments with different velocities are needed to confirm this idea. It should be noted that increased flow leads to a higher concentration of precipitated asphaltenes. In this case, the particles that pass through the pore increase at a higher flow rate, and the clogging intensifies. This is especially important during flow in a porous medium and the passage of particles through pores and cavities. Figure 6 shows the photos taken after the experiment at various flow rates. These images show more asphaltene deposits with a higher total flow rate. They also reveal more asphaltene deposits at the entrance, but fewer asphaltene deposits are observed further down the micromodel.

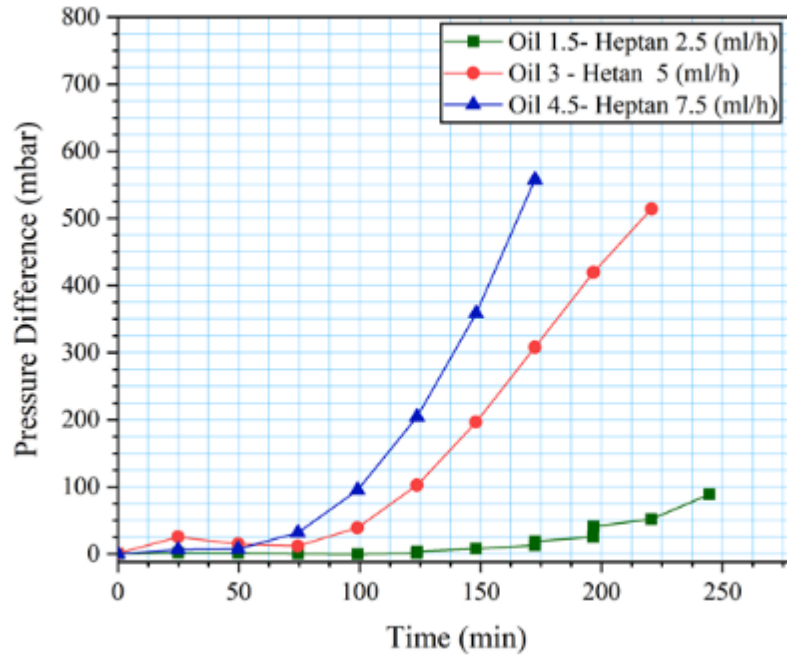


Figure 5

Pressure drop versus time for a constant n-heptane-to-oil ratio (62.5% v/v) at different volumetric flow rates of 4, 8, and 12 mL/h.

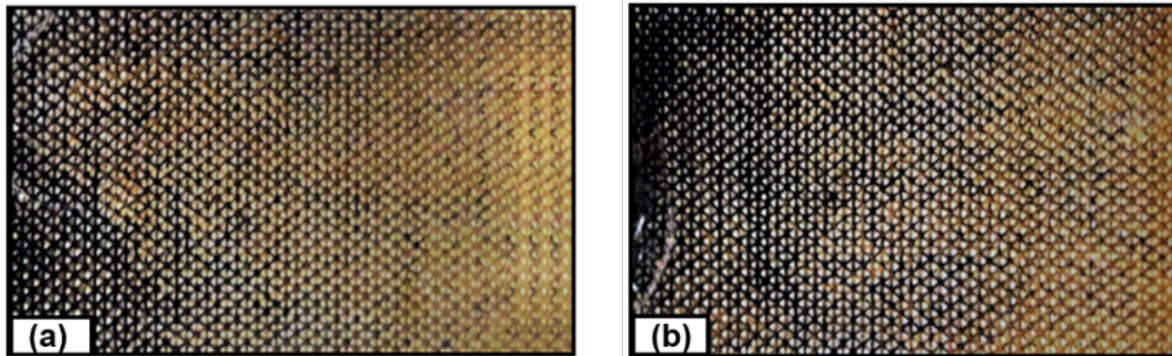
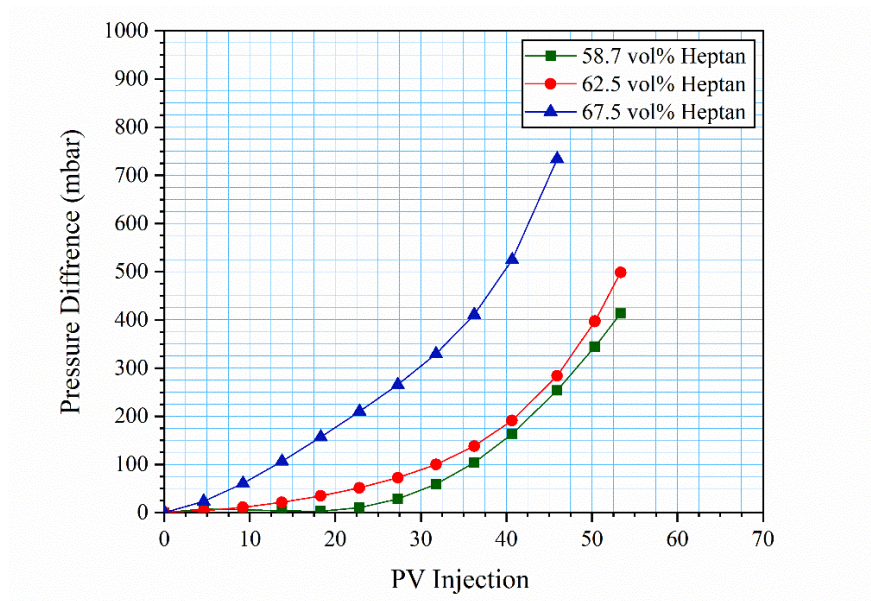


Figure 6

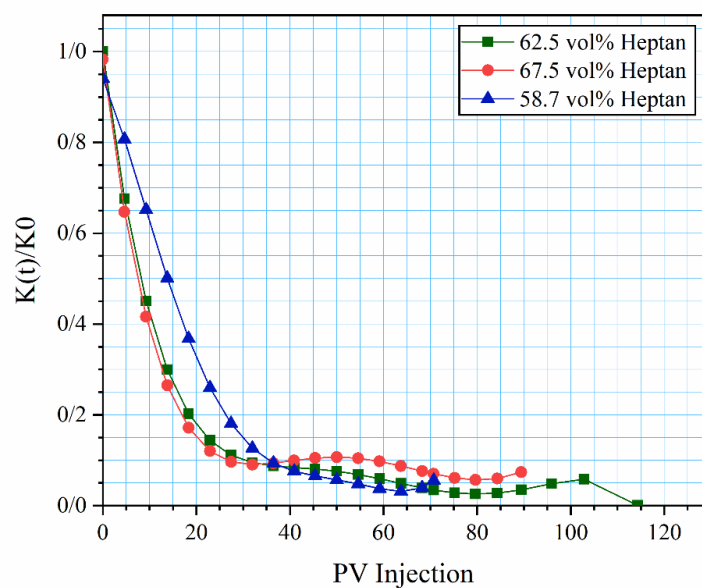
The images of glass micromodels at a flow rate of (a) 12 mL/h and (b) 8 mL/h two hours after the test, showing the progress of asphaltene deposits in the porous medium.

5.3. Effect of the precipitant fraction on asphaltene deposition

Figure 7 illustrates the pressure drop trend for three distinct experiments performed at a fixed volumetric flow rate of 8 mL/h with different n-heptane-to-crude oil ratios of 58.7%, 62.5%, and 67.5%. According to Figure 7, as the precipitant fraction increases, the pressure drop increases, proving a faster deposition rate. Increasing the content of n-heptane in the combination enhances both the initial precipitation kinetics and the concentration of asphaltenes in the fluid mixture. Equation (8) uses pressure drop data to generate the following graphic. Figure 8 describes the effect of the precipitant-to-oil volume ratio on permeability reduction. As seen in Figure 8, the precipitant-to-oil ratio increases the rate of permeability reduction because of the increasing instability of asphaltene.

**Figure 7**

Pressure drop versus pore volume injection for a constant flow rate of 8 mL/h at different n-heptane fractions of 58.7, 62.5, and 67.5 v/v %.

**Figure 8**

Permeability reduction versus pore volume injection for a constant flow rate of 8 mL/h at different n-heptane fractions of 58.7, 62.5, and 67.5 v/v %.

6. Comparison of experimental results with developed models

As the mechanisms of permeability impairment alter during the tests, it is essential to divide each test into discrete-time domains to determine dominant mechanisms. Consequently, the first test is divided into several domains based on the verification of experimental data with the proposed model, such that an exact match is seen for each period. Permeability data at 1.0–13.6 pore volumes domain are fitted using a quadratic polynomial function that accounts for the simultaneous coexistence of surface deposition and pore throat plugging mechanisms, as seen in Figure 9.

Figure 10 depicts the results of the fitting equation (Equation (13)) on permeability data through 13.6–35.1 injected pore volumes, which confirms that filtration cake formation is the predominant mechanism in this domain. Figure 11, allocated to the 35.1–57.2 pore volumes domain, shows that Equation (10) follows experimental data. Consequently, the pore throat-blocking mechanism is prominent.

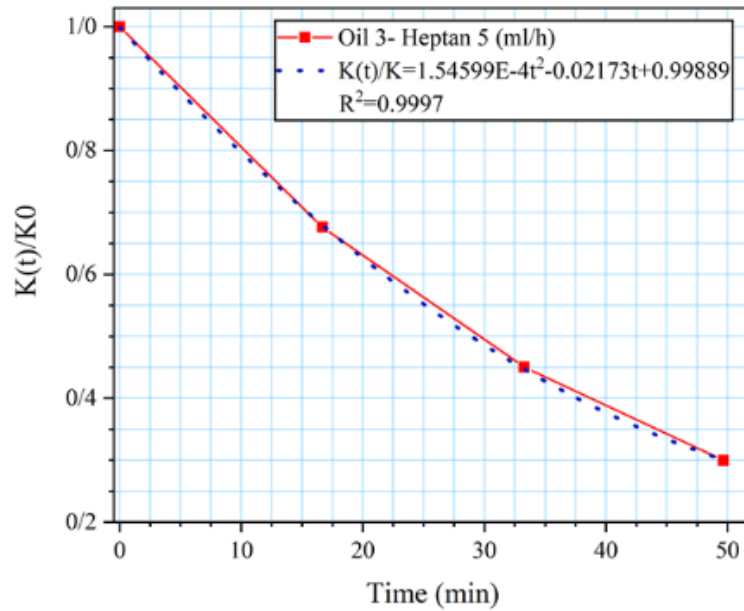


Figure 9

Regression of the experimental permeability data using a combination of surface deposition and pore throat plugging mechanisms in the 1.0–13.6 pore volumes domain.

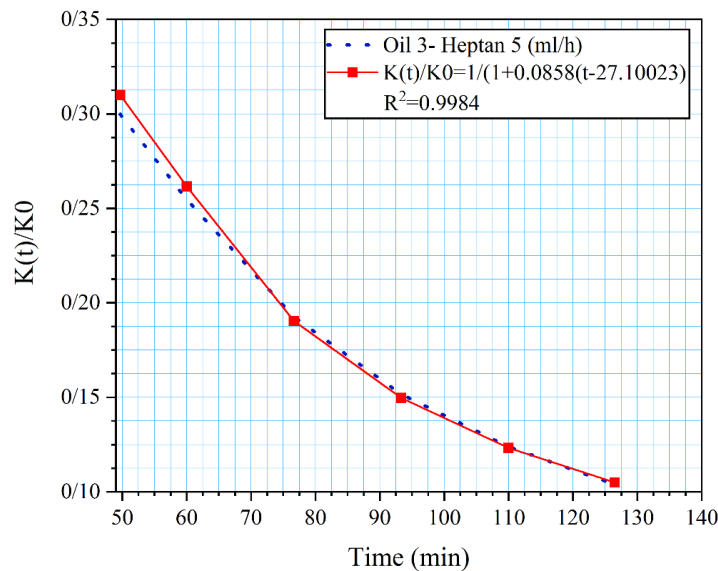


Figure 10

Regression of the experimental permeability data using filtration cake formation mechanism in the 13.6–35.1 pore volumes domain.

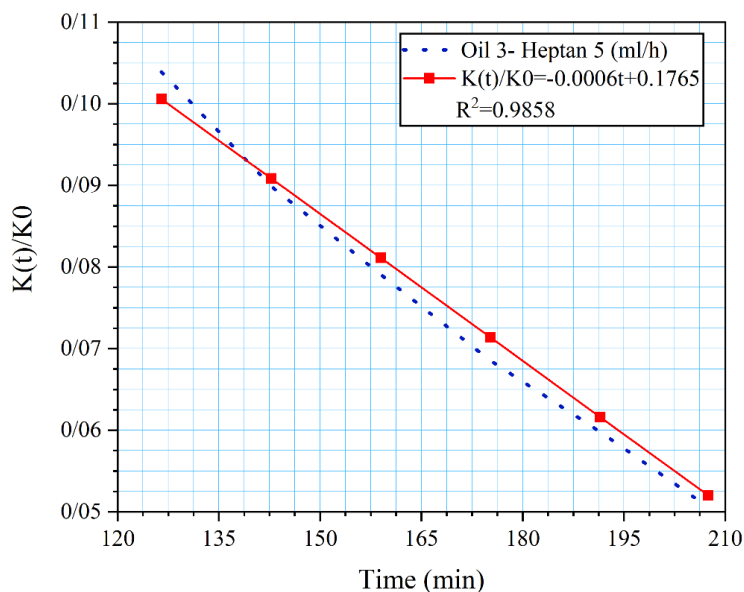


Figure 11

Regression of the experimental permeability data obtained using pore throat plugging mechanism in the 35.5–57.2 pore volume domain.

7. Conclusions

The following conclusions could be drawn from the above findings:

- The deposition rate increased with the flow velocity because, in the experiments performed and for the selected flow velocity, the stress rate was not high enough to overcome the kinetics of asphaltene deposition.
- The higher the precipitant-to-oil ratio, the higher the quantity of asphaltene deposition because the size and quantity of unstable asphaltene agglomerates grew with increasing the precipitant volume; in other words, asphaltene content had a direct influence on asphaltene deposition.
- As the precipitant carbon number increased, its compatibility with asphaltene rose; as a result, the deposition rate and instability of asphaltene decreased.
- Based on the validation of the experimental results with the proposed models of permeability reduction, a significant correlation was found between the collected data and the related models, indicating that the models might accurately predict permeability behavior.
- According to the examination of permeability impairment data, it could be shown that permeability reduction mechanisms fluctuated with pore volume.
- In the early stages of testing, the experimental data on permeability reduction were more in line with the quadratic polynomial function, accounting for the simultaneous coexistence of surface deposition and pore throat plugging mechanisms. In the middle of the experiment, the experimental data agreed with the filtration cake formation model. However, in the end, when this interval covered more time than the experiment, the experimental data were more consistent with the pore throat-plugging model.

Identifying the dominant mechanism in reducing permeability caused by asphaltene deposition could lead to developing chemical inhibitors or other preventative measures. This research can also be used to optimize asphaltene sedimentation processes.

Nomenclature

D	Diameter of the capillary
K	Permeability
L	Length of the tube
P	Friction pressure drop
μ	Fluid viscosity

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